

Code Listings

Project Title: Meal Planning and Recipe Management Web Application

Module: COM661 – Full-stack Strategies and Development

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1. Introduction

This document contains listings of the main files developed or modified for the Meal Planning and Recipe Management Web Application. Auto-generated Angular files and unmodified framework files have been excluded.

2. Included Files

- home component files
- recipe component files
- meal plan component files
- nutrition tracker component files
- login/authentication files
- Angular service files
- route configuration

3. Code Listings

Route Configuration

Src/app/components/home/home.html

```
<!-- HERO -->
<div class="hero">
  <div class="hero-overlay">
    <h1>Plan Your Meals Smarter 🍽️</h1>
    <p>Organise recipes, track nutrition, and simplify your weekly meals.</p>
  </div>
</div>
```

```

<!-- MAIN CARDS -->
<section class="preview-section">
  <div class="home-grid">

    <div class="card">
      <div class="card-icon">🍳</div>
      <h3>Recipes</h3>
      <p>Browse your recipe collection and manage saved recipes.</p>
      <a routerLink="/recipes" class="card-button">View Recipes</a>
    </div>

    <div class="card">
      <div class="card-icon">📅</div>
      <h3>Mealplans</h3>
      <p>View your weekly meal plans and planned meals.</p>
      <a routerLink="/mealplans" class="card-button">View Mealplans</a>
    </div>

    <div class="card">
      <div class="card-icon">+</div>
      <h3>Add Recipe</h3>
      <p>Create a new recipe with category and preparation details.</p>
      <a routerLink="/add-recipe" class="card-button">Add Recipe</a>
    </div>

    <div class="card">
      <div class="card-icon">📝</div>
      <h3>Add Mealplan</h3>
      <p>Create a weekly meal plan with multiple meals.</p>
      <a routerLink="/add-mealplan" class="card-button">Add Mealplan</a>
    </div>

  </div>
</section>

<!-- SUGGESTIONS + TRACKER -->
<section class="suggestions-section">

  <h2>High Protein Suggestions</h2>
  <p>Recipe ideas selected from your collection to support a balanced meal plan.</p>

```

```

<div class="suggestions-layout">

  <!-- LEFT: SUGGESTED RECIPES -->
  <div class="suggestions-grid" *ngIf="suggestedRecipes.length > 0">

    <div class="suggestion-card" *ngFor="let recipe of suggestedRecipes">

      <img
        [src]="recipe.image || 'https://images.unsplash.com/photo-1504674900247-0877df9cc836'"
        alt="Recipe image">

      <div class="suggestion-body">
        <h3>{{ recipe.name }}</h3>
        <p>{{ recipe.category }}</p>

        <a [routerLink]="['/recipes', recipe._id]">
          View Recipe
        </a>

        <div class="badge">High protein</div>
      </div>

    </div>

  </div>

  <!-- RIGHT: DYNAMIC TRACKER -->
  <aside class="tracker-panel">

    <!-- DYNAMIC CHART -->
    <div
      class="mini-chart"
      [style.background]="
        'conic-gradient(#16a34a 0% ' + proteinPercent + '%, ' +
        '#2563eb ' + proteinPercent + '% ' + (proteinPercent + carbsPercent) +
        '%, ' +
        '#f97316 ' + (proteinPercent + carbsPercent) + '% 100%)'
    ">
    </div>

    <h3>Nutrition Tracker</h3>

```

```

    <p>Estimate your daily calorie, protein and carbohydrate targets.</p>

    <!-- DYNAMIC STATS -->
    <div class="tracker-stats">

        <div>
            <strong>Protein</strong>
            <span>{{ proteinPercent }}%</span>
        </div>

        <div>
            <strong>Carbs</strong>
            <span>{{ carbsPercent }}%</span>
        </div>

        <div>
            <strong>Fats</strong>
            <span>{{ fatsPercent }}%</span>
        </div>

    </div>

    <a routerLink="/nutrition-tracker" class="tracker-btn">
        Open Tracker
    </a>

</aside>

</div>

</section>

```

Src/app/components/home/home.ts

```

import { Component, OnInit } from '@angular/core';
import { RouterLink } from '@angular/router';
import { CommonModule } from '@angular/common';
import { Api } from '../../../services/api';

@Component({
  selector: 'app-home',

```

```

imports: [RouterLink, CommonModule],
templateUrl: './home.html',
styleUrl: './home.css'
})
export class Home implements OnInit {
  suggestedRecipes: any[] = [];
  proteinPercent: number = 30;
  carbsPercent: number = 45;
  fatsPercent: number = 25;
  constructor(private api: Api) {}

  ngOnInit(): void {
    this.api.getRecipes().subscribe({
      next: (data: any) => {
        this.suggestedRecipes = data
          .filter((recipe: any) => Number(recipe.protein) >= 20)
          .slice(0, 3);

        if (this.suggestedRecipes.length === 0) {
          this.suggestedRecipes = data.slice(0, 3);
        },
      });
      const savedTracker = localStorage.getItem('trackerData');

      if (savedTracker) {
        const tracker = JSON.parse(savedTracker);

        const proteinCalories = tracker.protein * 4;
        const carbsCalories = tracker.carbs * 4;
        const fatCalories = tracker.fats * 9;

        const total = proteinCalories + carbsCalories + fatCalories;

        this.proteinPercent = Math.round((proteinCalories / total) * 100);
        this.carbsPercent = Math.round((carbsCalories / total) * 100);
        this.fatsPercent = Math.round((fatCalories / total) * 100);
      }
    }
  }
}

```

src/app/components/home/home.css

```
.hero {
  background-image:
    linear-gradient(rgba(17, 24, 39, 0.62), rgba(17, 24, 39, 0.62)),
    url('https://images.unsplash.com/photo-1490645935967-10de6ba17061');
  background-size: cover;
  background-position: center;
  padding: 60px 40px;
  border-radius: 18px;
  color: white;
  text-align: center;
  box-shadow: 0 12px 30px rgba(0,0,0,0.12);
}

.hero h1 {
  color: white;
  font-size: 42px;
  margin-bottom: 14px;
}

.hero p {
  color: #e5e7eb;
  font-size: 18px;
  max-width: 650px;
  margin: 0 auto 28px;
}

.main-btn,
.secondary-btn,
.card-button {
  text-decoration: none;
  padding: 12px 22px;
  border-radius: 8px;
  font-weight: 700;
  transition: all 0.25s ease;
}

.main-btn {
  background-color: #1f2937;
  color: white;
}

.main-btn:hover {
```

```
background-color: #111827;
transform: translateY(-2px);
}

.secondary-btn {
background-color: #e5e7eb;
color: #1f2937;
}

.secondary-btn:hover {
background-color: #d1d5db;
transform: translateY(-2px);
}

.preview-section {
margin-top: 30px;
padding: 30px 20px;
border-radius: 18px;

background: linear-gradient(
  135deg,
  #eef2f7,
  #e0e7ff
);

box-shadow: inset 0 1px 0 rgba(255,255,255,0.6);
}

.home-grid {
display: grid;
grid-template-columns: repeat(auto-fit, minmax(240px, 1fr));
gap: 20px;
}

.card {
background: white;
padding: 26px;
border-radius: 16px;
box-shadow: 0 8px 20px rgba(0,0,0,0.05);
transition: all 0.25s ease;
text-align: center;
}
```

```
.card:hover {  
  transform: translateY(-6px);  
  box-shadow: 0 14px 35px rgba(0,0,0,0.1);  
}
```

```
.card-icon {  
  font-size: 36px;  
  margin-bottom: 12px;  
}
```

```
.card h3 {  
  margin-bottom: 10px;  
  color: #1f2937;  
}
```

```
.card p {  
  color: #6b7280;  
  margin-bottom: 16px;  
}
```

```
.card-button {  
  display: inline-block;  
  padding: 10px 16px;  
  background: #1f2937;  
  color: white;  
  border-radius: 8px;  
  text-decoration: none;  
  font-weight: 600;  
}
```

```
.card-button:hover {  
  background: #111827;  
}
```

```
.home-grid {  
  display: grid;  
  grid-template-columns: repeat(2, 1fr);  
  gap: 28px;  
  max-width: 900px;  
  margin: 0 auto;  
}
```

```
@media (max-width: 700px) {  
  .home-grid {  
    grid-template-columns: 1fr;  
  }
```



```
}  
}  
.suggestions-section {  
  margin-top: 45px;  
  background: linear-gradient(135deg, #111827, #374151);  
  padding: 32px;  
  border-radius: 20px;  
  color: white;  
}  
  
.suggestions-section h2 {  
  margin-top: 0;  
  margin-bottom: 6px;  
  color: white;  
}  
  
.suggestions-section > p {  
  color: #d1d5db;  
  margin-bottom: 24px;  
}  
  
.suggestions-grid {  
  display: grid;  
  grid-template-columns: 1.2fr 1fr 1fr;  
  gap: 20px;  
}  
  
.suggestion-card {  
  background: white;  
  color: #1f2937;  
  border-radius: 16px;  
  overflow: hidden;  
  box-shadow: 0 12px 28px rgba(0,0,0,0.16);  
  transition: 0.25s ease;  
}  
  
.suggestion-card:first-child {  
  transform: scale(1.03);  
}  
  
.suggestion-card:hover {  
  transform: translateY(-6px);  
}
```

```
.suggestion-card img {
  width: 100%;
  height: 170px;
  object-fit: cover;
}

.suggestion-card:first-child img {
  height: 220px;
}

.suggestion-body {
  padding: 16px;
}

.suggestion-body h3 {
  margin: 0 0 6px;
  color: #111827;
}

.suggestion-body p {
  color: #6b7280;
  margin-bottom: 10px;
}

.suggestion-body::after {
  content: "High protein option";
  display: inline-block;
  margin-bottom: 12px;
  background: #dcfce7;
  color: #166534;
  padding: 5px 9px;
  border-radius: 999px;
  font-size: 12px;
  font-weight: 700;
}

.suggestion-body a {
  display: block;
  background: #1f2937;
  color: white;
  text-align: center;
  text-decoration: none;
}
```

```

padding: 10px 13px;
border-radius: 8px;
font-weight: 700;
}
.suggestions-layout {
display: grid;
grid-template-columns: 1fr 340px;
gap: 24px;
align-items: start;
}
@media (max-width: 850px) {
.suggestions-grid {
grid-template-columns: 1fr;
}

.suggestion-card:first-child {
transform: none;
}
}
.tracker-panel {
background: rgba(255, 255, 255, 0.12);
backdrop-filter: blur(12px);
border-radius: 22px;
padding: 26px;
box-shadow: 0 20px 40px rgba(0,0,0,0.25);
border: 1px solid rgba(255,255,255,0.15);
display: flex;
flex-direction: column;
align-items: center;
}

.mini-chart {
position: relative;
}

.mini-chart::before {
content: "Macros";
position: absolute;
inset: 0;
display: flex;
align-items: center;
justify-content: center;
font-weight: 800;

```

```
    color: #111827;
    font-size: 14px;
    z-index: 2;
}

.mini-chart::after {
    content: "";
    position: absolute;
    inset: 32px;
    background: white;
    border-radius: 50%;
}

.tracker-panel h3 {
    margin: 6px 0 8px;
    font-size: 26px;
    color: #111827;
}

.tracker-panel p {
    color: #6b7280;
    text-align: center;
    line-height: 1.5;
    max-width: 280px;
}

.tracker-stats {
    width: 100%;
    display: grid;
    gap: 12px;
    margin: 20px 0;
}

.tracker-stats div {
    display: flex;
    flex-direction: column;
    background: rgba(255,255,255,0.15);
    padding: 12px;
    border-radius: 12px;
}

.tracker-stats div::after {
    content: "";
    height: 6px;
```

```
margin-top: 8px;
border-radius: 6px;
background: linear-gradient(to right, #16a34a, #2563eb);
}

.tracker-stats strong {
  color: #111827;
}

.tracker-stats span {
  font-weight: 900;
  color: #059669;
}

.tracker-btn {
  width: 100%;
  background: #111827;
  color: white;
  text-decoration: none;
  text-align: center;
  padding: 13px;
  border-radius: 12px;
  font-weight: 800;
  transition: 0.25s ease;
}

.tracker-btn:hover {
  background: #000;
  transform: translateY(-2px);
}

.mini-chart {
  transition: transform 0.3s ease;
  cursor: pointer;
}

.mini-chart:hover {
  transform: scale(1.08) rotate(6deg);
}
```

Src/app/components/recipes/recipes.html

```
<section class="recipes-header">

  <h2>Recipes</h2>

  <p>Browse, search and manage your recipe collection.</p>
</section>
<div *ngIf="isLoading" class="loader">
  Loading recipes...
</div>

<div class="recipes-toolbar">
  <input
    type="text"
    [(ngModel)]="searchText"
    (input)="filterRecipes()"
    placeholder="Search by recipe name or category"
  />

  <span>{{ filteredRecipes.length }} recipes found</span>
</div>

<p class="message" *ngIf="message">{{ message }}</p>
<div class="recipes-grid" *ngIf="filteredRecipes.length > 0">
<div class="recipe-card" *ngFor="let recipe of filteredRecipes">

  <!-- IMAGE WRAPPER -->
  <div class="image-container">

    <img
      [src]="recipe.image || 'https://images.unsplash.com/photo-1504674900247-0877df9cc836'"
      class="recipe-img">

    <!-- OVERLAY -->
    <div class="overlay"></div>

    <!-- TITLE ON IMAGE -->
    <h3 class="image-title">
      {{ recipe.name }}
    </h3>

  </div>
</div>
</div>
```

```

</div>

<!-- DETAILS -->
<div class="card-body">

  <p><strong>Category:</strong> {{ recipe.category }}</p>
  <p><strong>Prep Time:</strong> {{ recipe.prep_time }} mins</p>

  <a class="view-btn" [routerLink]="['/recipes', recipe._id]">
    View Recipe
  </a>

</div>

</div>

```

Src/app/components/recipes/recipes.ts

```

import { Component, OnInit } from '@angular/core';
import { CommonModule } from '@angular/common';
import { FormsModule } from '@angular/forms';
import { RouterModule } from '@angular/router';
import { Api } from '../../services/api';

@Component({
  selector: 'app-recipes-list',
  standalone: true,
  imports: [CommonModule, FormsModule, RouterModule],
  templateUrl: './recipes-list.html',
  styleUrls: ['./recipes-list.css']
})
export class RecipesList implements OnInit {

  recipes: any[] = [];
  filteredRecipes: any[] = [];
  searchText: string = '';
  message: string = '';
  isLoading: boolean = true;

  constructor(private api: Api) {}

```

```

ngOnInit(): void {
  this.loadRecipes();
}

loadRecipes(): void {
  this.isLoading = true;

  this.api.getRecipes().subscribe({
    next: (data: any) => {
      this.recipes = data;
      this.filteredRecipes = data;
      this.isLoading = false;
    },
    error: () => {
      this.message = 'Could not load recipes';
      this.isLoading = false;
    }
  });
}

filterRecipes(): void {
  const search = this.searchText.toLowerCase();

  this.filteredRecipes = this.recipes.filter((recipe: any) =>
    recipe.name?.toLowerCase().includes(search) ||
    recipe.category?.toLowerCase().includes(search)
  );
}
}

```

src/app/components/recipes/recipes.css

```

.recipes-header {
  margin-bottom: 22px;
}

.recipes-header h2 {
  color: #1f2937;
  font-size: 34px;
  margin-bottom: 8px;
}

```



```
}

.recipes-header p {
  color: #6b7280;
  font-size: 16px;
}

.recipes-toolbar {
  display: flex;
  justify-content: space-between;
  align-items: center;
  gap: 14px;
  margin-bottom: 24px;
  flex-wrap: wrap;
}

.recipes-toolbar input {
  width: 100%;
  max-width: 420px;
  padding: 12px;
  border: 1px solid #d1d5db;
  border-radius: 8px;
}

.recipes-toolbar span {
  color: #4b5563;
  font-weight: 600;
}

.message {
  color: #1f2937;
  font-weight: 600;
  margin-bottom: 18px;
}

.recipes-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(260px, 1fr));
  gap: 22px;
}

.recipe-card {
  background: white;
  padding: 22px;
```

```
border-radius: 14px;
box-shadow: 0 8px 22px rgba(0,0,0,0.06);
transition: 0.25s ease;
}

.recipe-card:hover {
  transform: translateY(-4px);
  box-shadow: 0 12px 28px rgba(0,0,0,0.08);
}

.recipe-card h3 {
  margin-top: 0;
  color: #1f2937;
}

.recipe-card p {
  color: #4e5c6f;
}

.view-btn {
  display: inline-block;
  margin-top: 14px;
  background: #1f2937;
  color: white;
  padding: 10px 16px;
  border-radius: 8px;
  text-decoration: none;
  font-weight: 600;
}

.view-btn:hover {
  background: #111827;
}

.recipe-img {
  width: 100%;
  height: 180px;
  object-fit: cover;
  border-radius: 12px;
  margin-bottom: 14px;
}

.recipe-card {
  display: flex;
  flex-direction: column;
```

```
    justify-content: space-between;
}

.recipe-img {
  width: 100%;
  height: 180px;
  object-fit: cover;
  border-radius: 12px;
  margin-bottom: 14px;
}

.recipe-card {
  border-radius: 16px;
  overflow: hidden;
  background: white;
  box-shadow: 0 10px 25px rgba(0,0,0,0.06);
  transition: transform 0.25s ease;
}

.recipe-card:hover {
  transform: translateY(-6px);
}

.image-container {
  position: relative;
  height: 200px;
  overflow: hidden;
}

/* IMAGE */
.recipe-img {
  width: 100%;
  height: 100%;
  object-fit: cover;
  transition: transform 0.4s ease;
}

.recipe-card:hover .recipe-img {
  transform: scale(1.08);
}

.overlay {
```

```

    position: absolute;
    inset: 0;
    background: rgba(0,0,0,0.35);
}

.image-title {
    position: absolute;
    bottom: 12px;
    left: 14px;
    color: white;
    font-size: 20px;
    font-weight: 800;
    margin: 0;
}

.card-body {
    padding: 16px;
}

.card-body p {
    margin: 6px 0;
    color: #adb5c1;
}

.loader {
    text-align: center;
    padding: 30px;
    font-weight: 600;
    color: #6b7280;
}

```

src/app/components/meal-plans/meal-plans.html

```

<section class="mealplans-header">
  <h2>Mealplans</h2>
  <p>View and manage weekly meal plans created through the application.</p>
</section>

```

```

<div class="mealplans-toolbar">
  <span>{{ mealplans.length }} mealplans found</span>
</div>

<p class="message" *ngIf="message">{{ message }}</p>

<div class="mealplans-grid" *ngIf="mealplans.length > 0">
  <div class="mealplan-card" *ngFor="let mealplan of mealplans">
    <h3>{{ mealplan.user }}</h3>
    <p><strong>Week Start:</strong> {{ mealplan.week_start }}</p>

    <div class="meals-box" *ngIf="mealplan.meals && mealplan.meals.length > 0">
      <h4>Meals</h4>

      <ul>
        <li *ngFor="let meal of mealplan.meals">
          <span class="day">{{ meal.day }}</span>
          <span>{{ meal.meal_type }}</span>
          <span>{{ meal.recipe }}</span>
        </li>
      </ul>
      <button
        class="delete-btn"
        *ngIf="isAdmin()"
        (click)="onDelete(mealplan._id)">
        Delete Mealplan
      </button>
    </div>
  </div>
</div>

<p *ngIf="mealplans.length === 0">No mealplans available.</p>

```

src/app/components/meal-plans/meal-plans.ts

```

import { Component, OnInit } from '@angular/core';
import { CommonModule } from '@angular/common';
import { Api } from '../../../services/api';

```

```

@Component({
  selector: 'app-mealplans-list',
  imports: [CommonModule],
  templateUrl: './mealplans-list.html',
  styleUrls: ['./mealplans-list.css']
})
export class MealplansList implements OnInit {

  mealplans: any[] = [];
  message: string = '';

  isAdmin(): boolean {
    return localStorage.getItem('role') === 'admin';
  }

  constructor(private api: Api) {}

  ngOnInit(): void {
    this.loadMealplans();
  }

  loadMealplans(): void {
    this.api.getMealplans().subscribe({
      next: (data: any) => {

const username = localStorage.getItem('username');

this.mealplans = data.filter((mealplan: any) => {
  return mealplan.user === username;
});
},
    });
  }

  isLoggedIn(): boolean {
    return !!localStorage.getItem('token');
  }

  onDelete(id: string): void {
    this.api.deleteMealplan(id).subscribe({
      next: () => {
        this.message = 'Mealplan deleted successfully';
      }
    });
  }
}

```

```

        this.ngOnInit();
    },
    error: () => {
        this.message = 'Could not delete mealplan';
    }
  });
}
}

```

src/app/components/meal-plans/meal-plans.css

```

.mealplans-header {
  margin-bottom: 22px;
}

.mealplans-header h2 {
  color: #1f2937;
  font-size: 34px;
  margin-bottom: 8px;
}

.mealplans-header p {
  color: #6b7280;
  font-size: 16px;
}

.mealplans-toolbar {
  margin-bottom: 24px;
  color: #4b5563;
  font-weight: 600;
}

.message {
  color: #1f2937;
  font-weight: 600;
  margin-bottom: 18px;
}

.mealplans-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(290px, 1fr));
}

```

```
    gap: 22px;
}

.mealplan-card {
  background: white;
  padding: 24px;
  border-radius: 14px;
  box-shadow: 0 8px 22px rgba(0,0,0,0.06);
  transition: all 0.2s ease;
}

.mealplan-card:hover {
  transform: translateY(-4px);
  box-shadow: 0 12px 28px rgba(0,0,0,0.08);
}

.mealplan-card h3 {
  margin-top: 0;
  color: #1f2937;
  font-size: 24px;
}

.mealplan-card p {
  color: #4b5563;
}

.meals-box {
  margin-top: 18px;
  background: #f9fafb;
  padding: 14px;
  border-radius: 10px;
  border: 1px solid #e5e7eb;
}

.meals-box h4 {
  margin-top: 0;
  color: #1f2937;
}

.meals-box ul {
  padding: 0;
  margin: 0;
  list-style: none;
```



```

}

.meals-box li {
  display: grid;
  grid-template-columns: 1fr 1fr 1.5fr;
  gap: 8px;
  padding: 9px 0;
  border-bottom: 1px solid #e5e7eb;
  color: #4b5563;
}

.meals-box li:last-child {
  border-bottom: none;
}

.day {
  font-weight: 700;
  color: #1f2937;
}

.delete-btn {
  margin-top: 16px;
  background: #dc2626;
  color: white;
  border: none;
  padding: 10px 14px;
  border-radius: 7px;
  cursor: pointer;
  font-weight: 600;
}

.delete-btn:hover {
  background: #b91c1c;
}

```

src/app/components/nutrition-tracker/nutrition-tracker.html

```

<section class="tracker-page">
  <div class="tracker-form">
    <h2>Nutrition Tracker</h2>
    <p>Estimate your daily calorie and macronutrient intake targets.</p>

    <div class="form-group">

```

```

    <label>Age</label>
    <input type="number" [(ngModel)]="age" name="age">
  </div>

  <div class="form-group">
    <label>Weight (kg)</label>
    <input type="number" [(ngModel)]="weight" name="weight">
  </div>

  <div class="form-group">
    <label>Height (cm)</label>
    <input type="number" [(ngModel)]="height" name="height">
  </div>

  <div class="form-group">
    <label>Gender</label>
    <select [(ngModel)]="gender" name="gender">
      <option value="female">Female</option>
      <option value="male">Male</option>
    </select>
  </div>

  <div class="form-group">
    <label>Activity Level</label>
    <select [(ngModel)]="activity" name="activity">
      <option [ngValue]="1.2">Low activity</option>
      <option [ngValue]="1.55">Moderate activity</option>
      <option [ngValue]="1.9">High activity</option>
    </select>
  </div>

  <button (click)="calculate()">Calculate Targets</button>
</div>

<div class="tracker-results" *ngIf="calculated">
  <h2>Daily Targets</h2>

  <div class="chart"></div>

  <div class="result-grid">
    <div>
      <strong>{{ calories }}</strong>
      <span>Calories</span>
    </div>
  </div>
</div>

```

```

    </div>

    <div>
      <strong>{{ protein }}g</strong>
      <span>Protein</span>
    </div>

    <div>
      <strong>{{ carbs }}g</strong>
      <span>Carbs</span>
    </div>

    <div>
      <strong>{{ fats }}g</strong>
      <span>Fats</span>
    </div>
  </div>
</div>
</section>

```

src/app/components/nutrition-tracker/nutrition-tracker.ts

```

import { Component } from '@angular/core';
import { FormsModule } from '@angular/forms';
import { CommonModule } from '@angular/common';

@Component({
  selector: 'app-nutrition-tracker',
  imports: [FormsModule, CommonModule],
  templateUrl: './nutrition-tracker.html',
  styleUrls: ['./nutrition-tracker.css']
})
export class NutritionTracker {
  age: number = 0;
  weight: number = 0;
  height: number = 0;
  gender: string = 'female';
  activity: number = 1.2;

  calories: number = 0;
  protein: number = 0;
  carbs: number = 0;

```

```

fats: number = 0;

calculated: boolean = false;

calculate(): void {
  let bmr = 0;

  if (this.gender === 'male') {
    bmr = 10 * this.weight + 6.25 * this.height - 5 * this.age + 5;
  } else {
    bmr = 10 * this.weight + 6.25 * this.height - 5 * this.age - 161;
  }

  this.calories = Math.round(bmr * this.activity);
  this.protein = Math.round(this.weight * 1.6);
  this.carbs = Math.round((this.calories * 0.45) / 4);
  this.fats = Math.round((this.calories * 0.25) / 9);

  this.calculated = true;
const trackerData = {
  calories: this.calories,
  protein: this.protein,
  carbs: this.carbs,
  fats: this.fats
};

localStorage.setItem('trackerData', JSON.stringify(trackerData));

}
}

```

src/app/components/nutrition-tracker/nutrition-tracker.css

```

.tracker-page {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 28px;
  align-items: start;
}

.tracker-form,
.tracker-results {

```

```
background: white;
padding: 30px;
border-radius: 18px;
box-shadow: 0 10px 25px rgba(0,0,0,0.06);
}

.tracker-form h2,
.tracker-results h2 {
  margin-top: 0;
  color: #1f2937;
}

.tracker-form p {
  color: #6b7280;
}

.form-group {
  margin-bottom: 16px;
}

label {
  display: block;
  font-weight: 700;
  margin-bottom: 6px;
  color: #374151;
}

input,
select {
  width: 100%;
  padding: 11px;
  border: 1px solid #d1d5db;
  border-radius: 8px;
  box-sizing: border-box;
}

button {
  width: 100%;
  padding: 12px;
  background: #1f2937;
  color: white;
  border: none;
  border-radius: 8px;
}
```

```
font-weight: 800;
cursor: pointer;
}

.chart {
width: 210px;
height: 210px;
border-radius: 50%;
margin: 20px auto;
background: conic-gradient(
    #16a34a 0% 30%,
    #2563eb 30% 75%,
    #f97316 75% 100%
);
box-shadow: 0 10px 25px rgba(0,0,0,0.12);
}

.result-grid {
display: grid;
grid-template-columns: repeat(2, 1fr);
gap: 14px;
}

.result-grid div {
background: #f9fafb;
padding: 16px;
border-radius: 12px;
text-align: center;
}

.result-grid strong {
display: block;
font-size: 26px;
color: #1f2937;
}

.result-grid span {
color: #6b7280;
font-weight: 600;
}

@media (max-width: 850px) {
.tracker-page {
```

```
    grid-template-columns: 1fr;
  }
}
```

src/app/app.routes.ts

```
import { Routes } from '@angular/router';
import { Home } from '../components/home/home';
import { Login } from '../components/login/login';
import { RecipesList } from '../components/recipes-list/recipes-list';
import { MealplansList } from '../components/mealplans-list/mealplans-list';
import { AddRecipe } from '../components/add-recipe/add-recipe';
import { EditRecipe } from '../components/edit-recipe/edit-recipe';
import { AddMealplan } from '../components/add-mealplan/add-mealplan';
import { RecipeDetail } from '../components/recipe-detail/recipe-detail';
import { authGuard } from '../guards/auth-guard';
import { NutritionTracker } from '../components/nutrition-tracker/nutrition-tracker';

export const routes: Routes = [
  { path: '', component: Home, canActivate: [authGuard] },
  { path: 'login', component: Login },
  { path: 'recipes', component: RecipesList, canActivate: [authGuard] },
  { path: 'add-recipe', component: AddRecipe, canActivate: [authGuard] },
  { path: 'edit-recipe/:id', component: EditRecipe, canActivate: [authGuard] },
  { path: 'mealplans', component: MealplansList, canActivate: [authGuard] },
  { path: 'add-mealplan', component: AddMealplan, canActivate: [authGuard] },
  { path: 'recipes/:id', component: RecipeDetail, canActivate: [authGuard] },
  { path: 'nutrition-tracker', component: NutritionTracker, canActivate:
[authGuard] },
];
```

Src/app/components/login/login.html

```
<div class="login-container">

  <div class="login-info">
    <h1>Meal Planner</h1>
    <p>
      Plan meals smarter, manage recipes easily, and stay organised every week.
    </p>
```

```

<div class="feature-list">
  <div>🍳 Save and manage recipes</div>
  <div>📅 Create weekly meal plans</div>
  <div>🔒 Secure login options</div>
</div>
</div>

<div class="auth-card">

  <h2 *ngIf="!isSignup">Welcome Back</h2>
  <h2 *ngIf="isSignup">Create Account</h2>

  <p class="subtitle" *ngIf="!isSignup">
    Login to continue planning your meals.
  </p>

  <p class="subtitle" *ngIf="isSignup">
    Create an account to start using Meal Planner.
  </p>

  <form (ngSubmit)="onSubmit()">

    <div class="form-group">
      <label>Username</label>
      <input type="text" [(ngModel)]="username" name="username"
placeholder="Enter username">
    </div>

    <div class="form-group" *ngIf="isSignup">
      <label>Email</label>
      <input type="email" [(ngModel)]="email" name="email" placeholder="Enter
email">
    </div>

    <div class="form-group">
      <label>Password</label>
      <input type="password" [(ngModel)]="password" name="password"
placeholder="Enter password">
    </div>

    <button type="submit" class="primary-btn">
      {{ isSignup ? 'Sign Up' : 'Login' }}

```



```

    </button>

    <p class="message">{{ message }}</p>

    <p class="switch-text">
      <span (click)="switchMode()">
        {{ isSignup ? 'Already have an account? Login' : 'Need an account? Sign
Up' }}
      </span>
    </p>

  </form>

  <div class="divider">
    <span>or</span>
  </div>

  <button type="button" class="auth0-btn" (click)="loginAuth0()">
    Continue with Auth0
  </button>

</div>

</div>

```

Src/app/components/login/login.ts

```

import { Component, inject } from '@angular/core';
import { FormsModule } from '@angular/forms';
import { CommonModule } from '@angular/common';
import { Router } from '@angular/router';
import { Auth } from '../../../services/auth';
import { AuthService } from '@auth0/auth0-angular';

@Component({
  selector: 'app-login',
  imports: [FormsModule, CommonModule],
  templateUrl: './login.html',
  styleUrls: ['./login.css']
})
export class Login {
  isSignup: boolean = false;

```

```
username: string = '';
email: string = '';
password: string = '';
message: string = '';

auth0 = inject(AuthService);

constructor(private auth: Auth, private router: Router) {}

switchMode(): void {
  this.isSignup = !this.isSignup;
  this.message = '';
}

onSubmit(): void {
  if (this.isSignup) {
    const registerData = {
      username: this.username,
      email: this.email,
      password: this.password
    };

    this.auth.register(registerData).subscribe({
      next: () => {
        this.message = 'Sign up successful. You can now log in.';
        this.isSignup = false;
      },
      error: () => {
        this.message = 'Sign up failed';
      }
    });
  } else {
    const loginData = {
      username: this.username,
      password: this.password
    };

    this.auth.login(loginData).subscribe({
      next: () => {
        localStorage.setItem('username', this.username);
      }
    });
  }
}
```

```

    const adminUsers = ['admin', 'admin2'];

    if (adminUsers.includes(this.username.toLowerCase())) {
      localStorage.setItem('role', 'admin');
    } else {
      localStorage.setItem('role', 'user');
    }

    this.router.navigate(['/']);
  },
  error: () => {
    this.message = 'Login failed';
  }
});
}
}

loginAuth0(): void {
  this.auth0.loginWithRedirect();
}
}

```

src/app/components/login/login.css

```

.login-container {
  min-height: 75vh;
  display: grid;
  grid-template-columns: 1fr 420px;
  gap: 40px;
  align-items: center;
  max-width: 1100px;
  margin: 0 auto;
}

.login-info {
  background: linear-gradient(135deg, #111827, #374151);
  color: white;
  padding: 50px;
  border-radius: 22px;
  box-shadow: 0 14px 35px rgba(0, 0, 0, 0.12);
}

```

```
}

.login-info h1 {
  font-size: 48px;
  margin: 0 0 18px;
}

.login-info p {
  font-size: 18px;
  line-height: 1.6;
  color: #e5e7eb;
  max-width: 520px;
}

.feature-list {
  margin-top: 30px;
  display: grid;
  gap: 16px;
  font-size: 17px;
}

.feature-list div {
  background: rgba(255, 255, 255, 0.1);
  padding: 14px 16px;
  border-radius: 12px;
}

.auth-card {
  background: white;
  padding: 34px;
  border-radius: 20px;
  box-shadow: 0 14px 35px rgba(0, 0, 0, 0.10);
}

.auth-card h2 {
  text-align: center;
  margin: 0;
  color: #111827;
  font-size: 30px;
}

.subtitle {
  text-align: center;
```

```
    color: #6b7280;
    margin: 10px 0 26px;
}

.form-group {
    margin-bottom: 18px;
}

.form-group label {
    display: block;
    margin-bottom: 7px;
    font-weight: 700;
    color: #374151;
}

.form-group input {
    width: 100%;
    padding: 12px 13px;
    border: 1px solid #d1d5db;
    border-radius: 10px;
    box-sizing: border-box;
    font-size: 15px;
}

.form-group input:focus {
    outline: none;
    border-color: #111827;
    box-shadow: 0 0 0 3px rgba(17, 24, 39, 0.12);
}

.primary-btn,
.auth0-btn {
    width: 100%;
    padding: 13px;
    border: none;
    border-radius: 10px;
    font-weight: 800;
    cursor: pointer;
    font-size: 15px;
}

.primary-btn {
    background: #111827;
```

```
    color: white;
}

.primary-btn:hover {
    background: #000;
}

.auth0-btn {
    background: #f97316;
    color: white;
}

.auth0-btn:hover {
    background: #ea580c;
}

.message {
    text-align: center;
    margin-top: 14px;
    font-weight: 600;
    color: #374151;
}

.switch-text {
    text-align: center;
    color: #6b7280;
}

.switch-text span {
    cursor: pointer;
    color: #111827;
    font-weight: 800;
}

.switch-text span:hover {
    text-decoration: underline;
}

.divider {
    display: flex;
    align-items: center;
    margin: 26px 0;
    color: #9ca3af;
```

```

}

.divider::before,
.divider::after {
  content: "";
  flex: 1;
  height: 1px;
  background: #e5e7eb;
}

.divider span {
  padding: 0 12px;
}

@media (max-width: 850px) {
  .login-container {
    grid-template-columns: 1fr;
  }

  .login-info {
    text-align: center;
  }
}

```

Src/app/app.html

```

<nav class="topbar">
  <div class="logo">
    <ng-container *ngIf="auth0.isAuthenticated$ | async; else customUser">
      <ng-container *ngIf="auth0.user$ | async as user">
        Hello, {{ user.name || user.email }}
      </ng-container>
    </ng-container>

    <ng-template #customUser>
      <span *ngIf="isCustomLoggedIn(); else defaultTitle">
        Hello, {{ username }}
      </span>

      <ng-template #defaultTitle>
        Meal Planner
      </ng-template>
    </ng-template>
  </div>
</nav>

```

```

    </ng-template>
  </div>

  <div class="nav-links">
    <a routerLink="/">Home</a>
    <a routerLink="/login" *ngIf="!(auth0.isAuthenticated$ | async)
    && !isCustomLoggedIn()">Login</a>
  </div>

  <div class="nav-actions">
    <button
      class="logout-btn"
      *ngIf="auth0.isAuthenticated$ | async"
      (click)="auth0.logout({ logoutParams: { returnTo:
window.location.origin } })">
      Logout
    </button>

    <button
      class="logout-btn"
      *ngIf="!(auth0.isAuthenticated$ | async) && isCustomLoggedIn()"
      (click)="logoutCustom()">
      Logout
    </button>
  </div>
</nav>

<div class="page-container">
  <router-outlet></router-outlet>
</div>

<footer class="footer">
  Manage your meal plans easily @2026.
</footer>

```

Src/app/app.ts

```

import { Component, DoCheck, inject } from '@angular/core';
import { RouterOutlet, RouterLink, Router } from '@angular/router';
import { CommonModule } from '@angular/common';
import { Auth } from '../services/auth';
import { AuthService } from '@auth0/auth0-angular';

```



```

@Component({
  selector: 'app-root',
  imports: [RouterOutlet, RouterLink, CommonModule],
  templateUrl: './app.html',
  styleUrls: ['./app.css']
})
export class App implements DoCheck {
  username: string = '';
  protected readonly window = window;

  auth0 = inject(AuthService);

  constructor(private auth: Auth, private router: Router) {}

  ngDoCheck(): void {
    this.username = localStorage.getItem('username') || '';
  }

  isCustomLoggedIn(): boolean {
    return !!localStorage.getItem('token');
  }

  logoutCustom(): void {
    this.auth.logout();
    localStorage.removeItem('username');
    localStorage.removeItem('role');
    this.router.navigate(['/']);
  }
}

```

src/app/app.css

```

body {
  margin: 0;
  font-family: Arial, sans-serif;
  background: linear-gradient(to bottom, #f8fafc, #eef2f7);
  color: #222;
}

.topbar {
  background-color: #111827;

```

```
    color: white;
    display: flex;
    justify-content: space-between;
    align-items: center;
    padding: 18px 34px;
    box-shadow: 0 6px 20px rgba(0,0,0,0.08);
}

.logo {
    font-size: 22px;
    font-weight: 700;
    letter-spacing: 0.3px;
}

.nav-links {
    display: flex;
    gap: 22px;
    align-items: center;
}

.nav-links a {
    color: white;
    text-decoration: none;
    font-weight: 500;
    padding: 8px 12px;
    border-radius: 6px;
    transition: all 0.2s ease;
}

.nav-links a:hover {
    background: rgba(255,255,255,0.12);
}

.logout-btn {
    background-color: #dc2626;
    color: white;
    border: none;
    padding: 9px 14px;
    border-radius: 6px;
    cursor: pointer;
    font-weight: 600;
    transition: all 0.2s ease;
}
```

```

.logout-btn:hover {
  background-color: #b91c1c;
  transform: translateY(-1px);
}

.page-container {
  padding: 40px;
  min-height: 80vh;
}

.footer {
  text-align: center;
  padding: 24px;
  color: #9ca3af;
  font-size: 14px;
}

```

Src/app/services/auth.service.ts

```

import { Injectable } from '@angular/core';
import { HttpClient } from '@angular/common/http';
import { Observable, tap } from 'rxjs';

@Injectable({
  providedIn: 'root'
})
export class Auth {
  private baseUrl = 'http://127.0.0.1:5001';

  constructor(private http: HttpClient) {}

  login(data: any): Observable<any> {
    return this.http.post(`${this.baseUrl}/login`, data).pipe(
      tap((response: any) => {
        if (response.token) {
          localStorage.setItem('token', response.token);
        }
      })
    );
  }
}

```

```
register(data: any): Observable<any> {  
    return this.http.post(`${this.baseUrl}/register`, data);  
}  
  
logout(): void {  
    localStorage.removeItem('token');  
}  
  
getToken(): string | null {  
    return localStorage.getItem('token');  
}  
}
```